BrandCheck

Pre-publication brand governance: multimodal content review, source-grounded AI, and accountable human decisions.

ONE-MINUTE JUDGE JOURNEY (NO SETUP REQUIRED)

1. Open /example for a zero-login review. 2. Sign in at /workspace for shared roles. 3. Import PDF/DOCX at /documents for page or paragraph findings. 4. Submit a revision; a separate reviewer decides.

HOSTED WORKING PRODUCT

https://brandcheck-codestar-judge.freelynx19.chatgpt.site/workspace

PUBLIC SOURCE REPOSITORY

https://github.com/Manas-Jain-24/brandcheck

DEPLOYED PRODUCT INTERFACE: SHARED REVIEW & VISUAL LEADS

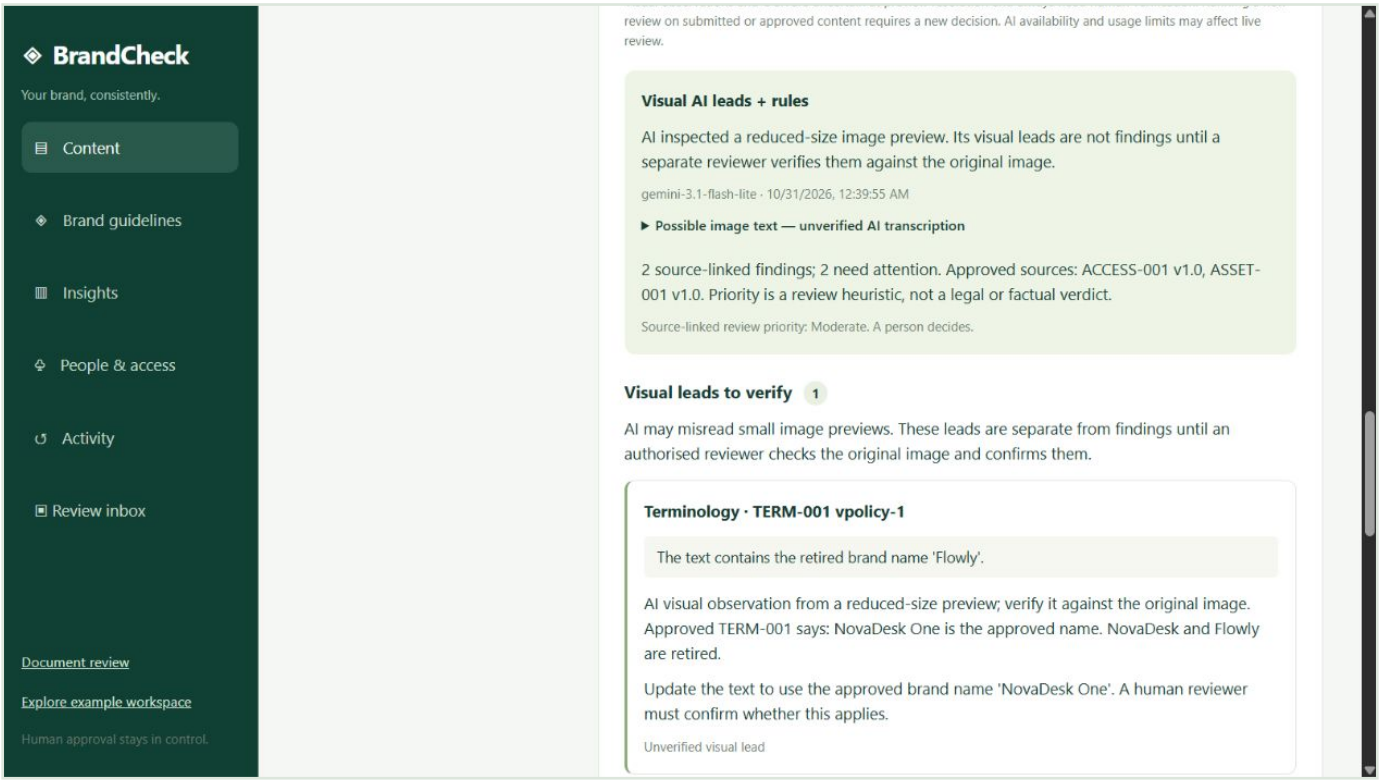


Figure 1: Deployed multi-tenant workspace showing deterministic findings, grounded Gemini review, and unverified visual leads awaiting reviewer confirmation.

01 / DELIVERED ROUND 3 IMPROVEMENTS

Execution of Promised Capabilities

Our Round 2 submission promised three major upgrades: grounded semantic review, real shared review, and PDF/DOCX document intake. The implementation below follows those commitments. Visual governance and policy synchronization are additional capabilities.

IMPROVEMENT 1: GROUNDED SEMANTIC REVIEW

Delivered: Consented Gemini analysis checks tone and claims against active approved sources. Model findings are accepted only when their source IDs are valid and excerpts appear in the draft. The result cites the source, labels uncertainty and offers a suggested rewrite while leaving the decision with a person. Deterministic rules remain available when AI is unavailable.
Scope: Internal source grounding does not independently verify third-party facts on the live web.

IMPROVEMENT 2: REAL SHARED REVIEW

Delivered: Platform-authenticated sessions, Cloudflare D1 persistence and server-enforced Admin, Creator and Reviewer roles. A creator submits an exact revision; a separate reviewer comments, requests changes or approves. Self-approval is rejected by the server. Optimistic revision checks prevent stale decisions, and audit events preserve source versions and decisions.
Scope: Collaboration uses an in-app inbox and manual sharing; external email or Slack notifications are not implemented.

IMPROVEMENT 3: PDF/DOCX DOCUMENT INTAKE

Delivered: Client-side PDF.js and Mammoth/fflate extraction for text-based PDF and DOCX, plus TXT and Markdown. Findings retain PDF page or Word paragraph locations. The document view supports editing extracted sections and handing a shortened draft into the shared workspace for review and approval.
Scope: Scanned pages have no OCR, page locations are section anchors rather than pixel boxes, and workspace drafts are limited to 10,000 characters.



Additional capability: public GitHub policy import and atomic policy regression recheck. This production run shows affected drafts returning to 'needs review'.

Transactional Integrity & Verified Tests

Rather than treating AI as a black box, BrandCheck pairs serverless edge performance with strict schema validation, optimistic concurrency control, and deterministic rule fallbacks.

FULL-STACK ARCHITECTURE & DATA FLOW

Edge Runtime & UI: React 19, TypeScript, Vinext/Vite and Tailwind CSS running on Cloudflare Workers.
Persistence & Schema: Cloudflare D1 (serverless SQLite) with additive Drizzle ORM migrations. Zero runtime table creation.
Grounded AI Pipeline: Browser → Same-Origin API → Private Runtime Secret → Google Gemini 3.1. Draft excerpts must physically exist in submitted text and map to an active approved source. Unmapped model claims are filtered out.
Safe Ingestion: Client-side PDF.js worker; Mammoth with fflate zip-expansion limits; 160 KB API request streaming guard.

CONCURRENCY & ANTI-SELF-REVIEW ENGINEERING

- Atomic Compare-and-Swap (CAS): Draft updates and review decisions enforce exact revision matching (version = version + 1 WHERE version = ?). Concurrent mutations return HTTP 409 Conflict without data corruption.
- Anti-Self-Review Guard: The server verifies draft.author_id !== user.userId on every review decision and visual lead confirmation. Creators cannot rubber-stamp their own content via UI or direct API manipulation.
- Graceful AI Fallback & Quotas: High-volume quota allowance (180 requests/day, 20/10-min IP bucket). When AI is unreachable or exhausted, deterministic term/alt-text/claim rules execute seamlessly with clear provenance labeling.

VERIFICATION RECORD (46 AUTOMATED TESTS PASSING)

- SQLite Service Suite: 46 comprehensive service/helper tests pass locally using actual D1 migrations and in-memory SQLite.
- Tested Scenarios: Workspace isolation, role boundaries, self-review denial, reviewer assignment, CAS concurrent writes, transaction rollback, policy regression, document location mapping, and visual lead confirmation.
- Real Document Fixtures: Production validation tested with actual 2-page PDF and zipped XML DOCX files.
- Strict Type Safety: Strict TypeScript build passed with zero errors (tsc --noEmit).

GOVERNANCE TOOLING & ENTERPRISE FEATURES

Includes an exportable Brand Clearance Certificate, a live Pre-Flight Status Matrix, 1-click Batch Terminology Corrections, Reviewer Decision Templates, and a smart WCAG 2.2 Contrast Optimizer with auto-adjusting compliant color calculation.

Boundaries, Roadmap & Contributions

We believe in absolute intellectual honesty. The capabilities described in this report represent verified, functional software. Features outside our MVP scope, along with our development roadmap, are documented below.

WHAT DOES NOT WORK YET (DELIBERATE MVP BOUNDARIES)

- Document Extraction: Requires extractable text. Scanned images have no OCR; files are bounded to 15 MB, 40 PDF pages, and 80k chars. Shared drafts support up to 10k plain text characters without rich layout retention.
- Visual Governance: Exact SHA-256 fingerprint matching, alt-text presence, text contrast ratio, and dominant palette heuristics are supported. Full-image logo clearance, trademark placement, and canvas coordinate boxes are not implemented.
- Policy Sync and Audit: Supports on-demand public GitHub JSON import with admin activation. Automated polling, private OAuth (Notion/Confluence), and NLP memo conversion remain roadmap items. Audit log is append-only D1 SQLite, not SOC2-certified.

PRODUCT ROADMAP AND FUTURE DEVELOPMENT PLAN

Phase 1: Ingestion & CMS Connectors (Q1): Native Google Docs and MS Word 365 add-ins for inline guideline checks; Notion and Confluence bi-directional webhooks; serverless OCR fallback for scanned PDF documents.
Phase 2: Enterprise Governance & Workflows (Q2): Multi-tier approval routing (Copywriter → Brand Director → Legal Counsel); real-time Slack/Teams alerts; background cron monitoring for claim evidence expiration.
Phase 3: Continuous Compliance & Automation (Q3): Headless CI/CD brand linting CLI (brandcheck-cli) for GitHub Actions; cryptographically signed audit ledger exports for compliance audits.

THIRD-PARTY REUSE AND AI ASSISTANCE DISCLOSURES

Reused Components: Foundation utilizes the Sites Vinext/Vite starter on Cloudflare Workers. Parsing incorporates PDF.js (Apache-2.0), Mammoth (BSD-2-Clause), and fflate (MIT).
AI Models and Assistance: Semantic review calls Google Gemini 3.1 Flash Lite with server credentials. Significant AI coding assistance was utilized for rapid prototyping, refactoring, and test authoring.

VERIFIED TEAM MEMBER CONTRIBUTIONS

Manas Jain (Team Leader): Product architecture, Cloudflare Workers & D1 transactional backend, Zod validation schemas, optimistic concurrency (CAS), Gemini API integration, and edge deployment.

Mayank Kumar (Teammate): Client-side document intake engine (PDF.js / Mammoth), WCAG 2.2 contrast auto-fix algorithm, UI layout components, test fixture authoring, and documentation.

OFFICIAL BUILD WINDOW DECLARATION

Repository history confirms first commit on 26 October 2026, 1:49 AM IST. All claimed features were developed inside the official build window. Deployed globally on Cloudflare Workers.